PySpark Functions Cheatsheet

Column & Row Operations

Function	Description	Example
withColumn()	Add/modify a column	df.withColumn("new", df["col"] + 1)
select()	Select specific columns	df.select("col1", "col2")
filter() / where()	Filter rows	df.filter(df["age"] > 30)
drop()	Drop a column	df.drop("col")
dropDuplicates()	Remove duplicates	df.dropDuplicates(["col"])
distinct()	Unique rows	df.distinct()
withColumnRenamed()	Rename column	df.withColumnRenamed("old", "new")
orderBy() / sort()	Sort rows	df.orderBy("col", ascending=False)
cache() / persist()	Store DataFrame in memory/disk	df.cache()

Join & Combine

Function	Description	Example
join()	Join two DataFrames	df1.join(df2, "id", "inner")
union()	Append DataFrames	df1.union(df2)
unionByName()	Append by matching column names	df1.unionByName(df2)
repartition()	Increase partitions	df.repartition(10)
coalesce()	Reduce partitions	df.coalesce(1)

Aggregation & Grouping

Function	Description	Example
groupBy()	Group by column(s)	df.groupBy("col")
agg()	Multiple aggregations	df.agg(F.max("col"), F.min("col"))
describe()	Summary stats	df.describe().show()
collect_list()	List aggregation	collect_list("name")
collect_set()	Set (distinct) aggregation	collect_set("name")

Null & Conditional Handling

Function	Description	Example
fillna()	Replace nulls	df.fillna(0)
isNull()	Check nulls	df.filter(df["col"].isNull())
when()	If-else logic	when(df.age < 18, "minor").otherwise("adult")
lit()	Add literal value	lit("USA")

Function	Description	Example
row_number()	Row number per window	row_number().over(Window.partitionBy("col"))
rank()	Rank with gaps	rank().over(Window.partitionBy("col"))

dense_rank()	Rank without gaps	dense_rank().over(Window.partitionBy("col"))

Window & Ranking Functions

Date & Time

Function	Description	Example
current_date()	Today's date	current_date()
current_timestamp()	Current timestamp	current_timestamp()
to_date()	Convert to date	to_date(df["ts"])
date_format()	Format date	date_format("date", "yyyy-MM")

Complex Types & JSON

Function	Description	Example
explode()	Flatten arrays	explode("skills")
array() / struct()	Create complex types	array("c1", "c2"), struct("c1", "c2")
get_json_object()	Extract JSON field	get_json_object("json_col", "$.name")
from_json() / to_json()	Parse or stringify JSON	from_json("col", schema)